



NATIONAL TEACHERS COLLEGE

SDG 4: Open-Access Resource Tracker

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Introduction:

The project, Open access resource tracker, is designed to support students, teachers, and self-learners by providing a centralized platform where users can easily find free and accessible educational resources online. These resources may include e-books, research articles, video lessons, learning websites, tutorials, and downloadable study materials from trusted educational platforms. The system aims to organize and track these resources to make learning more convenient, efficient, and accessible for everyone.

This project aligns with SDG 4: Quality Education, one of the United Nations Sustainable Development Goals that focuses on ensuring inclusive and equitable quality education and promoting lifelong learning opportunities for all. Many learners struggle to access educational materials due to financial limitations, lack of awareness, or difficulty finding reliable sources online. Through the Open Access Resource Tracker, users can quickly discover learning materials without needing to spend money on expensive books or subscriptions.

The chosen SDG target for this project is Target 4.5, which aims to eliminate inequalities in education and ensure equal access to learning opportunities for vulnerable and disadvantaged groups. By providing free and open educational resources in one platform, the project helps reduce barriers to education and supports students from different backgrounds in achieving better learning opportunities.

Overall, the Open Access Resource Tracker promotes accessible education, digital learning, and equal opportunities, helping contribute to achievement of SDG 4: Quality Education.

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Requirement Analysis:

Functional Requirements

Authentication

- The system shall allow users to log in using username, password, and role.
- The system shall validate credentials and display specific error messages for wrong username or password.
- The system shall support two roles: Administrator and Standard User.
- The system shall redirect users to the main dashboard upon successful login

Resource Management

- The system shall allow Admin to add a new resource with Title, Author, URL, and Category.
- The system shall require Title and URL when adding a resource.
- The system shall allow Admin to delete an existing resource with a confirmation prompt
- The system shall display all resources in a grid with columns: ID, Title, Author, URL, and Category.

Category Management

- The system shall allow Admin to view all existing categories.
- The system shall allow Admin to add new categories
- The system shall link each resource to a category via CategoryID.

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Search & Filter

- The system shall allow users to search resources by Title, Author, or Category.
- The system shall filter the grid in real-time as the user type.

Role-Based Access Control

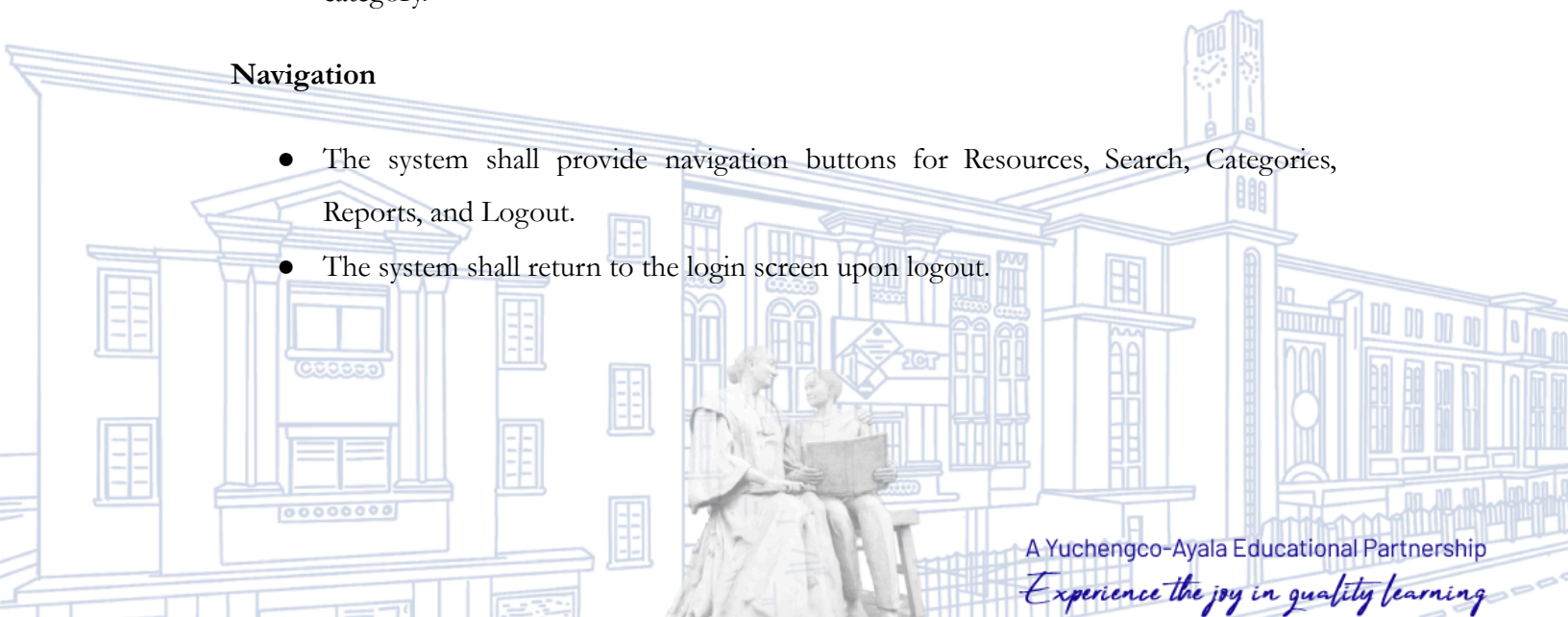
- The system shall restrict Add, Edit, Delete, and Categories to Admin only.
- The system shall display an “Access Denied” warning when a Standard User attempts to access restricted features.
- The system shall allow both roles to view resources and generate reports.

Report Generation

- The system shall generate a Monthly Resource Consumption report showing Categories, Total Resources, Added This Month, and Most Accessed.
- The system shall allow users to print the report.
- The system shall allow users to export the report as a CSV file.
- The system shall allow users to click a category row to view all resources under that category.

Navigation

- The system shall provide navigation buttons for Resources, Search, Categories, Reports, and Logout.
- The system shall return to the login screen upon logout.



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Non-Functional Requirements

Security

- The system shall enforce role-based access so Standard User cannot perform admin actions.
- The system shall not expose raw database errors to the user.

Performance

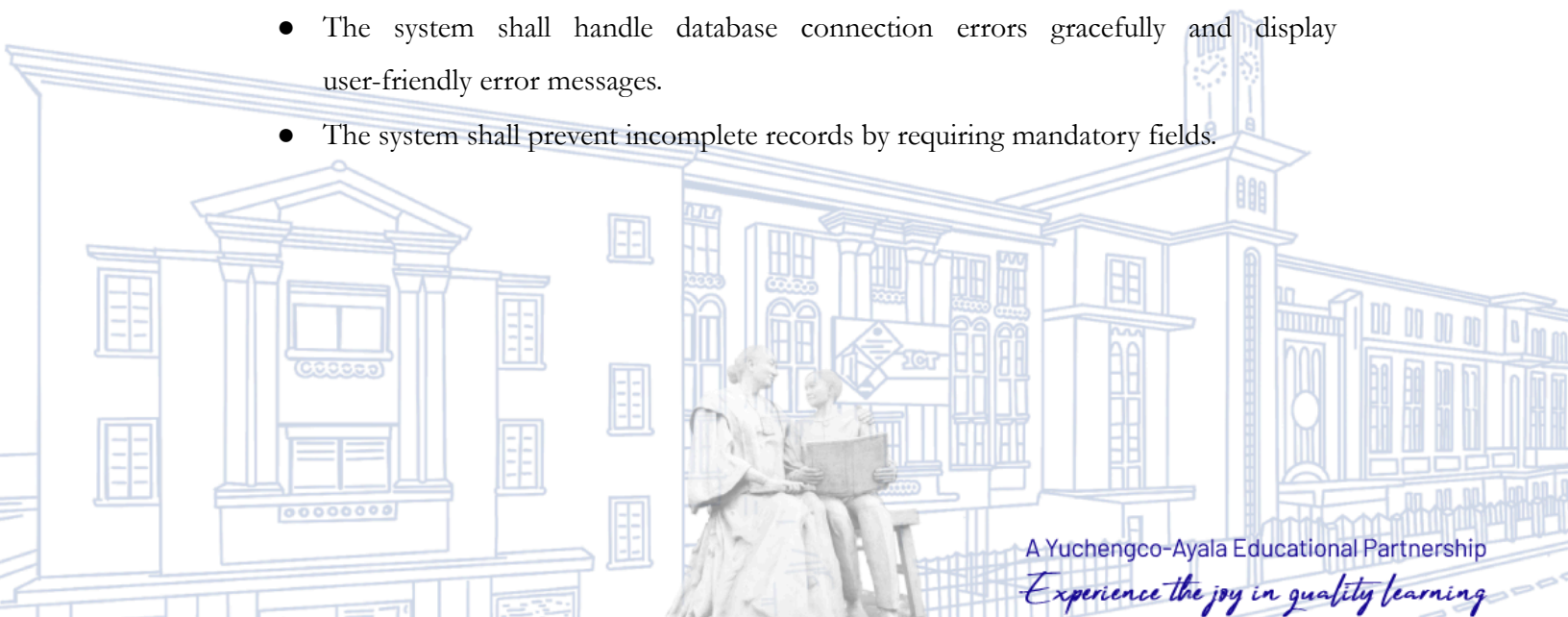
- The system shall load resources without timeout using unlimited command timeout.
- The system shall respond to search input in real-time without noticeable delay.

Usability

- The system shall display clear error and success messages for all user actions.
- The system shall prompt for confirmation before deleting a resource.
- The system shall show available categories when adding or editing a resource.

Reliability

- The system shall handle database connection errors gracefully and display user-friendly error messages.
- The system shall prevent incomplete records by requiring mandatory fields.



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Maintainability

- The system shall use a centralized dbConnection class for all database connections
- The system shall separate concerns across forms: frmLogin, frmMain, and frmReports.

Compatibility

- The system shall run on windows using .Net Windows Forms
- The system shall connect to a Microsoft SQL Server Database.



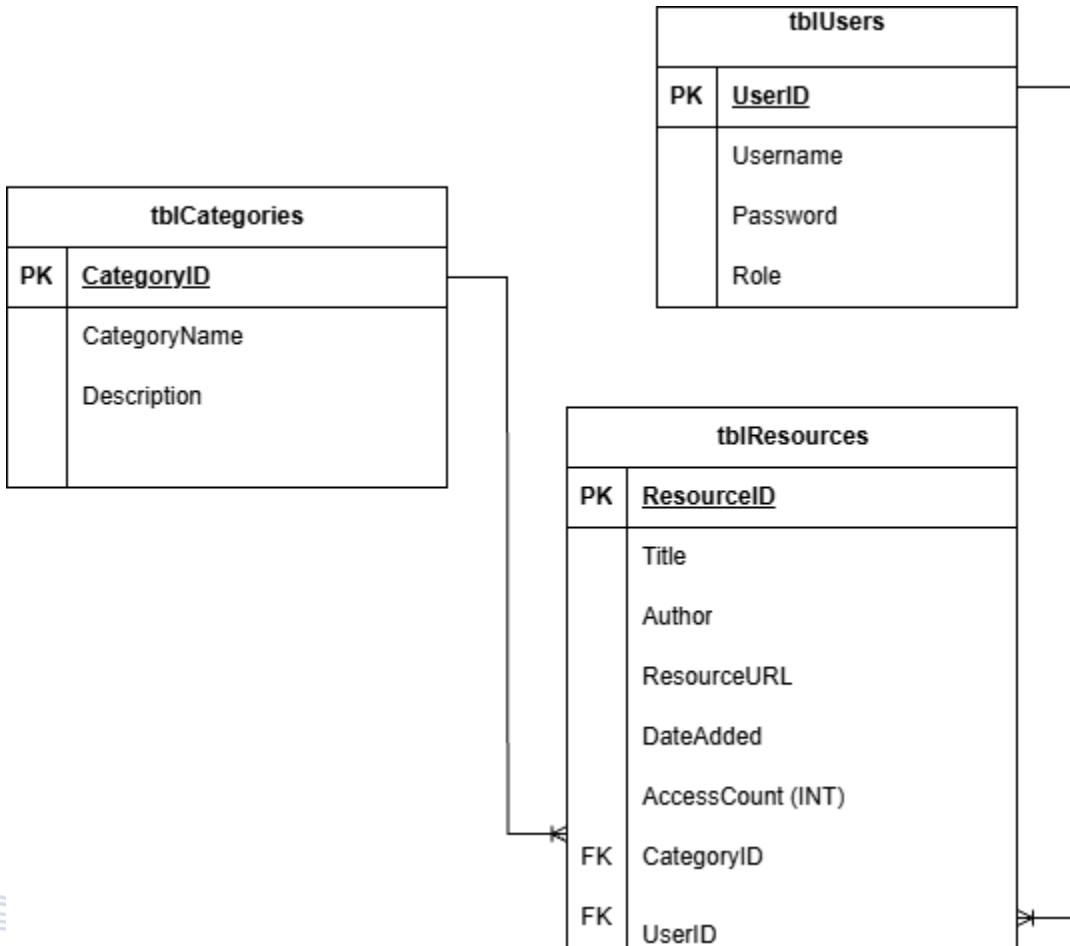
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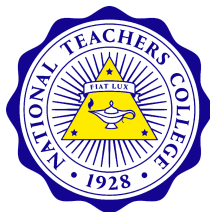
Database Design:

ERD



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Data Dictionary

Table 1: tblCategories

Field Name	Data Type	Constraint	Description
CategoryID	INT	Primary key, Identity(1,1)	Unique Identifier for each category
CategoryName	NVARCHAR(100)	Not null	Name of the category
Description	NVARCHAR(255)	Null	Optional description of the category

Table 2: tblResources

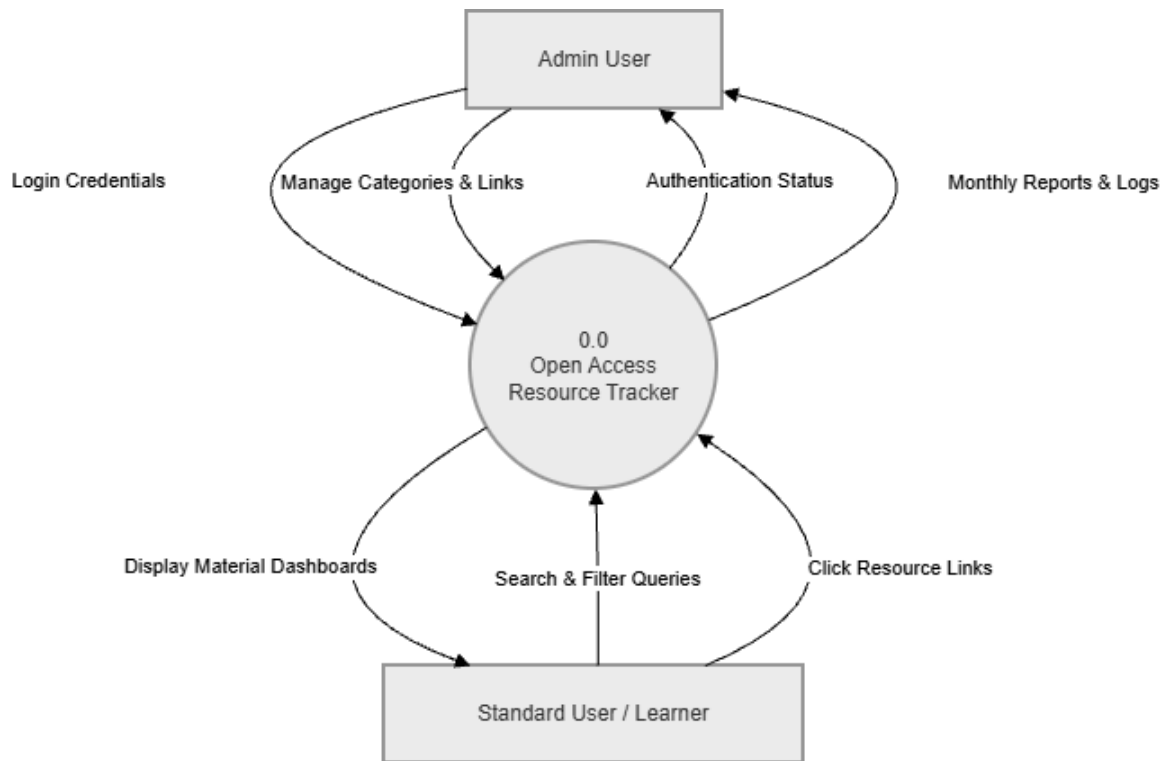
Field Name	Data Type	Constraint	Description
ResourceID	INT	Primary key, Identity(1,1)	Unique identifier for each category
Title	NVARCHAR(200)	Not null	Title of the resource
Author	NVARCHAR(100)	Null	Author of the resource
ResourceURL	NVARCHAR(MAX)	Not null	URL link to the resource
DateAdded	DATETIME	Default GETDATE()	Date and time the resource was added
CategoryID	INT	Foreign key →tblCategories(CategoryID)	Links resource to a category



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System Architecture:

DFD Level 0:

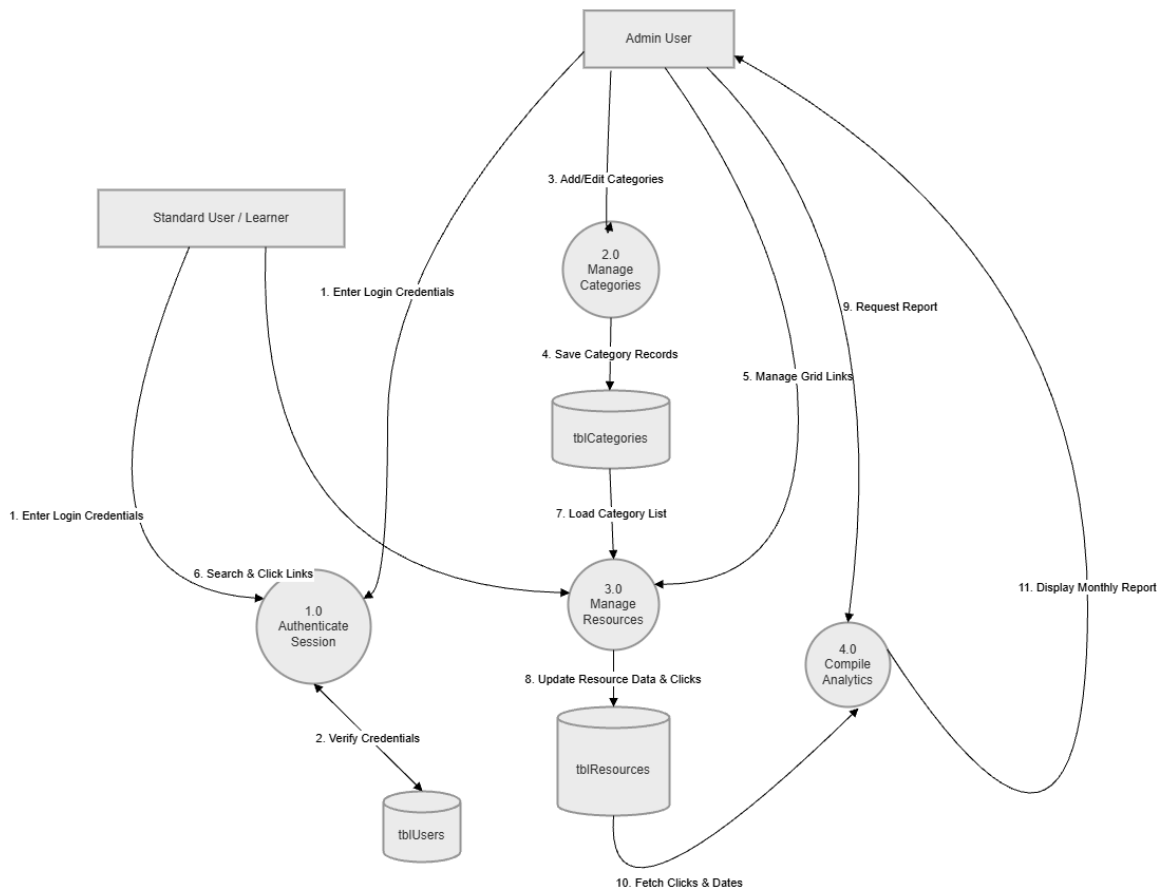


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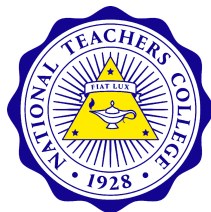


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DFD Level 1:



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Individual Contributions:

Module/Form	Description	Member Description
frmLogin	Login form, role-based authentication, user validation	Sean Lawrence Tapangan
frmMain	Main dashboard, navigation, role access control	Sean Lawrence Tapangan, John Michael Tolentino, Charles Gatus, Gershon Martinez, Rheigj Vergel Castro
dbConnection	Database connection, connection string setup	Sean Lawrence Tapangan, John Michael Tolentino
tblResources	Database table design for resources	Charles Gatus, Gershon Martinez
tblCategories	Database table design for categories	Rheigj Vergel Castro, Charles Gatus, Gershon Martinez
Add Resources	InputBox flow, INSERT query, category selection	Sean Lawrence Tapangan, John Michael Tolentino
Edit Resources	Row selection, UPDATE query, category editing	Charles Gatus, Sean Lawrence Tapangan
Delete Resources	Row selection, DELETE query, confirmation dialog	Rheigj Vergel Castro, Gershon Martinez
Search/Filter	Real-time search by title, author, category	Sean Lawrence Tapangan, John Michael Tolentino, Gershon Martinez



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frmReports	Report generation, print, export to CSV, drilldown	John Michael Tolentino
Roles-Based Access	Admin VS Standard User button restrictions, SetRole method	Sean Lawrence Tapangan, John Michael Tolentino



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